MADHAV INSTITUTE OF TECHNOLOGY & SCIENCE, GWALIOR (M.P.), INDIA
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DEPARTMENT OF COMPUTER SCIENCE & DESIGN • CORNERSTONE PROJECT (29243110)

Development of an AI-Enabled Civic Issue Reporting and Resolution Management System

CORNERSTONE PROJECT MILESTONE 4 PROGRESS & OPERATIONAL INTEGRATION REPORT

MILESTONE 4 STATUS
82%
Complete

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EVALUATION STAGE
Milestone 4 Review
Session: July–Dec 2026

CODE REPOSITORY
`main (Commit d0c5cac)`
Date: October 12, 2026

SYSTEM INTEGRATION	BACKEND API READY	FRONTEND WORKSPACES	MONGOOSE MODELS
82%	**88%**	**80%**	**95%**
End-to-end operational flow	22+ verified REST routes	Citizen, Dept & Admin views	8 Collections & Seeders

1.0 Executive Summary & Problem Statement

Municipal complaint handling mechanisms in urban centers predominantly rely on fragmented walk-in or manual procedures, resulting in delayed prioritization, lack of citizen visibility, unverified closures, and record duplication. The **AI-Enabled Civic Issue Reporting and Resolution Management System** delivers a unified, role-governed web application connecting citizens, departmental response teams (Road Maintenance, Sanitation, Water Works, Street Lighting), and administrative leadership. In this **Milestone 4** phase, the platform has completed the core lifecycle: **multi-role authentication & Google OAuth 2.0 with role onboarding**, **automated sequential complaint identification (`SC-2026-XXXX`)**, **AI-driven heuristic classification & urgency scoring**, **multipart media evidence ingestion**, **department SLA computation**, and **in-app status change notifications**.

2.0 Modular System Architecture

Client-Server Microservice Architecture

1. PRESENTATION TIER
React 19 • Vite • Tailwind CSS
Role-gated dashboards for Citizens, Department Staff, and Admins. Protected routing, session timer & Recharts.

2. APPLICATION API TIER
Node.js • Express • Multer • JWT
Stateless REST APIs, role authorization, sequential ID generation (`SC-2026-XXXX`), and audit logging.

3. AI TRIAGE ENGINE
NLP Keywords • Severity Scorer
Natural-language category prediction, emergency hazard detection (wires, floods), and discrepancy flags.

4. DATA PERSISTENCE
MongoDB Atlas • Mongoose ODM
Compound indexed documents for complaints, status timelines, notifications, departments, and seeders.

3.0 Milestone 4 Accomplishments & Project Scope

✅ Key Milestone 4 Accomplishments

- **Dual Auth with Role Gating:** JWT email/password auth plus Google OAuth with interactive role onboarding modal.
- **Complaint Lifecycle Pipeline:** Submission → Under Review → Assigned → In Progress → Resolved with citizen rating (1–5★).
- **AI Severity & Discrepancy Detection:** Urgency keyword weighting and automatic flagging when citizen category diverges from AI prediction.
- **Multipart Proof Ingestion:** Multer upload validating 5MB size limits, JPEG/PNG/WEBP MIME types, and before/after photo proof.

🎯 Final Sprint Deliverables

- **Interactive GIS Hotspot Canvas:** Live MongoDB coordinates rendered to cluster pins on Google Maps / Leaflet.
- **Trained Machine Learning Model:** Transitioning from heuristic engine to TF-IDF / Transformer & CNN microservice.
- **Aggregated Ward Analytics:** MongoDB aggregation pipeline for monthly resolution curves and ward density distribution.
- **Admin User Governance:** Global user control table (`/api/users`) for account auditing and suspension.

4.0 Comprehensive Module Progress Breakdown

Verified against live codebase & test suites

#	Module / Subsystem	Backend	Frontend	Overall	Status	Implementation Details & Codebase Artifacts
1	**Authentication & RBAC**	95%	95%	**95%**	COMPLETED	JWT stateless auth (`citizen` , `department` , `admin`), Google OAuth token verification, role selection modal, and session timeout handlers.
2	**Complaint Lifecycle**	92%	90%	**91%**	COMPLETED	Sequential tracking ID generator (`SC-2026-XXXX`), category auto-router, 5-stage status lifecycle, chronological audit trail, and citizen 1–5★ rating submission.
3	**AI Triage & Severity Scorer**	85%	85%	**85%**	COMPLETED	Natural-language category prediction, hazard keyword urgency assessment (floods, live wires), and discrepancy flag when citizen input differs from AI prediction.
4	**Media & Photo Ingestion**	90%	90%	**90%**	COMPLETED	Multer storage engine with 5MB validation, MIME filtering (JPEG/PNG/WEBP), submission attachment previews, and resolution verification photo evidence.
5	**In-App Notification Hub**	85%	85%	**85%**	COMPLETED	Event-driven triggers on status mutation, real-time unread badge counter, `/api/notifications` feed, and mark-as-read individual/bulk endpoints.
6	**Department Operations**	85%	75%	**80%**	COMPLETED	Real-time SLA resolution calculation API, department officer portal, issue inspection modal, and resolution comment logs. Admin reassignment UI integrated.
7	**GIS Mapping & Hotspots**	65%	60%	**62%**	IN PROGRESS	Latitude and Longitude schema fields persisted on all complaints; Google Maps / Leaflet component canvas created. Live database geo-pin binding in progress.
8	**Admin User Governance**	60%	60%	**60%**	IN PROGRESS	Schemas exist for Citizen, Admin, and Department collections; Frontend tables created. Global `/api/users` CRUD and suspension toggles in progress.
9	**Analytics & Ward Metrics**	60%	55%	**58%**	IN PROGRESS	High-level KPI statistics endpoint active (`/api/complaints/stats`); Recharts visual templates designed. MongoDB aggregation pipelines for ward trends in progress.

5.0 Core Technical Capabilities Implemented in Milestone 4

1. Auto-Sequenced Complaint Lifecycle `complaintController.js`

Issues filed through the API are assigned atomic, human-readable IDs formatted as `SC-YYYY-XXXX` . The system parses issue categories (e.g., Pothole, Water Leakage, Garbage) and automatically assigns jurisdiction to the responsible municipal department. Each state update generates an immutable timeline entry recording timestamp, actor, status delta, and remarks.

2. Dual Authentication with Role Onboarding `authController.js`

The platform provides standard JWT-based email/password authentication backed by bcrypt hashing alongside Google OAuth 2.0. Users authenticating via Google OAuth are routed through a dedicated onboarding modal (`GoogleRoleModal.jsx`) enabling explicit selection between Citizen, Department Officer, or Administrator privileges before session token creation.

3. AI Classification & Discrepancy Detection `NLP Urgency Engine`

Evaluates complaint natural-language text against category lexicons and hazard keywords (e.g., exposed wires, severe flooding near schools). Computes priority levels (Low, Medium, High, Critical). If citizen manual selection conflicts with AI prediction, the discrepancy is automatically flagged for administrative review.

4. Evidence Ingestion & Inspection Flow `uploadMiddleware.js`

Implemented a robust file upload middleware powered by Multer with strict disk storage limits (5MB) and MIME-type filtering. Supports initial citizen photo submission during complaint creation as well as post-resolution proof photo attachments uploaded by municipal officers during the closure transition.

6.0 Operational REST API Endpoint Inventory

22+ Endpoints Active, Authenticated & Verified

Method	Route Path	Access Level	Functional Purpose & Response Payload
POST	`/api/auth/register`	Public	Registers new user account with hashed password and role assignment.
POST	`/api/auth/login`	Public	Authenticates user credentials and returns signed JWT access token.
POST	`/api/auth/google`	Public	Verifies Google OAuth ID token and binds user profile with role onboarding.
GET	`/api/auth/me`	Authenticated	Fetches current authenticated profile from User, Citizen, or Department schema.
GET	`/api/complaints`	Role-Scoped	Paginated complaint registry filtered automatically by user role and department.
POST	`/api/complaints`	Citizen	Ingests new complaint with location, category, and optional Multer photo upload.
GET	`/api/complaints/my`	Citizen	Returns complaint history filed by the logged-in citizen with status breakdown.
GET	`/api/complaints/stats`	Authenticated	Computes live aggregated metrics: total, pending, in-progress, resolved, and SLA %.
GET	`/api/complaints/:id`	Authenticated	Retrieves complete complaint entity including audit trail and feedback details.
PUT	`/api/complaints/:id/status`	Dept / Admin	Updates status pipeline with remarks and resolution verification photo evidence.
PUT	`/api/complaints/:id/assign`	Admin Only	Reassigns complaint to a different municipal department with audit entry.
POST	`/api/complaints/:id/feedback`	Citizen	Records citizen resolution satisfaction rating (1–5 stars) and review comments.
GET	`/api/departments`	Authenticated	Lists municipal departments along with dynamic real-time resolution metrics.
GET	`/api/notifications`	Authenticated	Retrieves chronologically sorted in-app alert notifications for current user.
PUT	`/api/notifications/read-all`	Authenticated	Updates all unread notifications for active user to read status simultaneously.

7.0 Edge-Case Hardening & Failure Recovery Mechanisms

Fault-Tolerant System Engineering

⚡ AI Engine Downtime

If classification microservice times out, the submission is saved with manual category and flagged as `pending_classification`. A background retry job triages without blocking the citizen.

📍 Missing Location Access

If browser GPS is blocked, system avoids corrupting database with dummy (0,0) coordinates. Accepts manual address while keeping coordinates null to protect duplicate geo-clustering.

🔁 Rapid Double-Submission

Debounce guards and client-side lockouts detect rapid duplicate button taps, binding the user to the existing registered ticket instead of inserting duplicate database records.

8.0 Database Architecture & Schema Collections

MongoDB / Mongoose ODM Schemas

📦 Complaint Entity Schema

- `complaintId` : Unique sequential string (`SC-2026-0001`).
- `title, description, category` : Indexed civic descriptors.
- `status` : Enum [`Submitted` , `Under Review` , `Assigned` , `In Progress` , `Resolved` , `Closed`].
- `location` : Ward, address, and GeoJSON coordinates (`lat/lng`).
- `images, resolutionImages` : Paths to verified evidence files.
- `timeline[]` : Chronological audit transitions with actors and remarks.
- `feedback` : Rating (1-5★), review text, and citizen submission date.

👥 Supporting Collections & Relations

- `User` : Base identity with email, bcrypt password hash, and role type.
- `Citizen` : Profile extended with ward, phone number, address, and history.
- `Department` : Name, department code, nodal officer, and SLA targets.
- `Notification` : Recipient reference, complaint link, unread boolean flag.
- `Setting & SupportTicket` : Configuration baselines and helpdesk tickets.
- `seed.js` : Automated database initialization for test and demo data.

9.0 Milestone Schedule & Phased Implementation Timeline

Milestone 4: Operational Integration Phase

MILESTONE 1: COMPLETE	MILESTONE 2: COMPLETE	MILESTONE 3: COMPLETE	MILESTONE 4: CURRENT
Architecture & Models Mongoose schemas, Express base routing, and modular architecture.	**Authentication & Security** JWT verification, Google OAuth 2.0 onboarding modal, and RBAC.	**Lifecycle & Media** Complaint auto-IDs, Multer storage, and status notification hub.	**Department & Workflows** SLA metrics, status modals, feedback ratings, and integration testing.

10.0 Milestone 4 Quality Assurance & Verification Matrix

Verified through automated and manual test routines

Subsystem Test Scope	Verification Result	Observed Behavior & Test Assessment
Authentication & RBAC	PASSED (100%)	Token validation checks enforce route isolation. Google OAuth role modal cleanly assigns Citizen/Dept/Admin credentials.
Complaint Ingestion & ID	PASSED (100%)	Sequential IDs (`SC-2026-XXXX`) generate deterministically without collisions. Automatic department category mapping verified.
Multer Media Ingestion	PASSED (100%)	Files exceeding 5MB or non-image MIME types are rejected with clean 400 error codes. Valid images persist to `/uploads`.
Notification Dispatcher	PASSED (100%)	Status changes trigger asynchronous notification documents. Unread badge reflects active count and toggles upon bulk read.
Department SLA Calculation	PASSED (100%)	Dynamic resolution percentage and active complaint breakdown computed accurately across all municipal departments.

11.0 Machine Learning Transition & Dataset Baseline Plan

Roadmap for Heuristic-to-Trained Model Evolution

ML Phase	Target Objective & Technical Activity	Duration	Success Benchmark Metric
Phase 1	Curate and label evaluation dataset (350+ examples, 50 per category); compute baseline accuracy of heuristic engine.	2 Weeks	Ground-truth labeled test split across 7 civic categories.
Phase 2	Feature engineering & model exploration (TF-IDF + Classical ML vs Lightweight Transformer text classifier + MobileNet CNN for photos).	2 Weeks	Category Accuracy & Severity Score within ±1 level tolerance.
Phase 3	Model training, validation, and head-to-head comparison against rule-based baseline on held-out split.	2 Weeks	Trained model exceeds heuristic baseline by >15% accuracy.
Phase 4	Staged FastAPI microservice deployment behind feature flag with rule-based engine retained as fallback.	1 Week	Zero-downtime inference with fallback latency <200ms.

12.0 Final Phase Deliverables & Technical Risk Mitigations

📌 Deliverables for Final Submission

- **Interactive GIS Map:** Bind live MongoDB complaint coordinates into `ComplaintMap.jsx` with cluster pins and department filters.
- **AI Duplicate Detection Service:** Fast-API Python microservice using vector embeddings to flag duplicate civic complaints.
- **Admin User Governance:** Global user control table (`/api/users`) to inspect, suspend, or update roles.
- **Ward Analytics Aggregation:** MongoDB pipeline generating monthly resolution charts and ward-level issue distribution.

🛡️ Technical Risks & Architectural Mitigations

- **Risk: Spatial Query Latency on Map Pins**
  Mitigation: 2dsphere indexing on MongoDB coordinates combined with viewport bounding-box query limits.
- **Risk: High Volume Submission Collisions**
  Mitigation: Atomic counter sequences and optimistic retry handlers guarantee ID uniqueness.
- **Risk: Storage Overhead from Proof Photos**
  Mitigation: Strict 5MB Multer caps with sharp image optimization pipeline scheduled before cloud backup.
- **Risk: Unhandled Client Token Expiration**
  Mitigation: Centralized Axios/Fetch interceptor triggers automated logout and session modal prompt.